The Fundamental Isomorphism ϕ

$$\phi : \mathbb{F}_2[x]/\langle x^7 - 1\rangle \xrightarrow{\sim} \mathbb{F}_2^7$$

$$\sum_{i=0}^{6} a_i x^i \longmapsto (a_0, a_1, a_2, a_3, a_4, a_5, a_6)$$

Abstract Algebra

Message: $m(x) = m_0 + m_1 x + m_2 x^2 + m_3 x^3$

Generator: $g(x) = \mathbf{1} + \mathbf{1}x + \mathbf{0}x^2 + \mathbf{1}x^3$

Codeword Generation:

$$c(x) = m(x)g(x) \pmod{x^7 - 1}$$

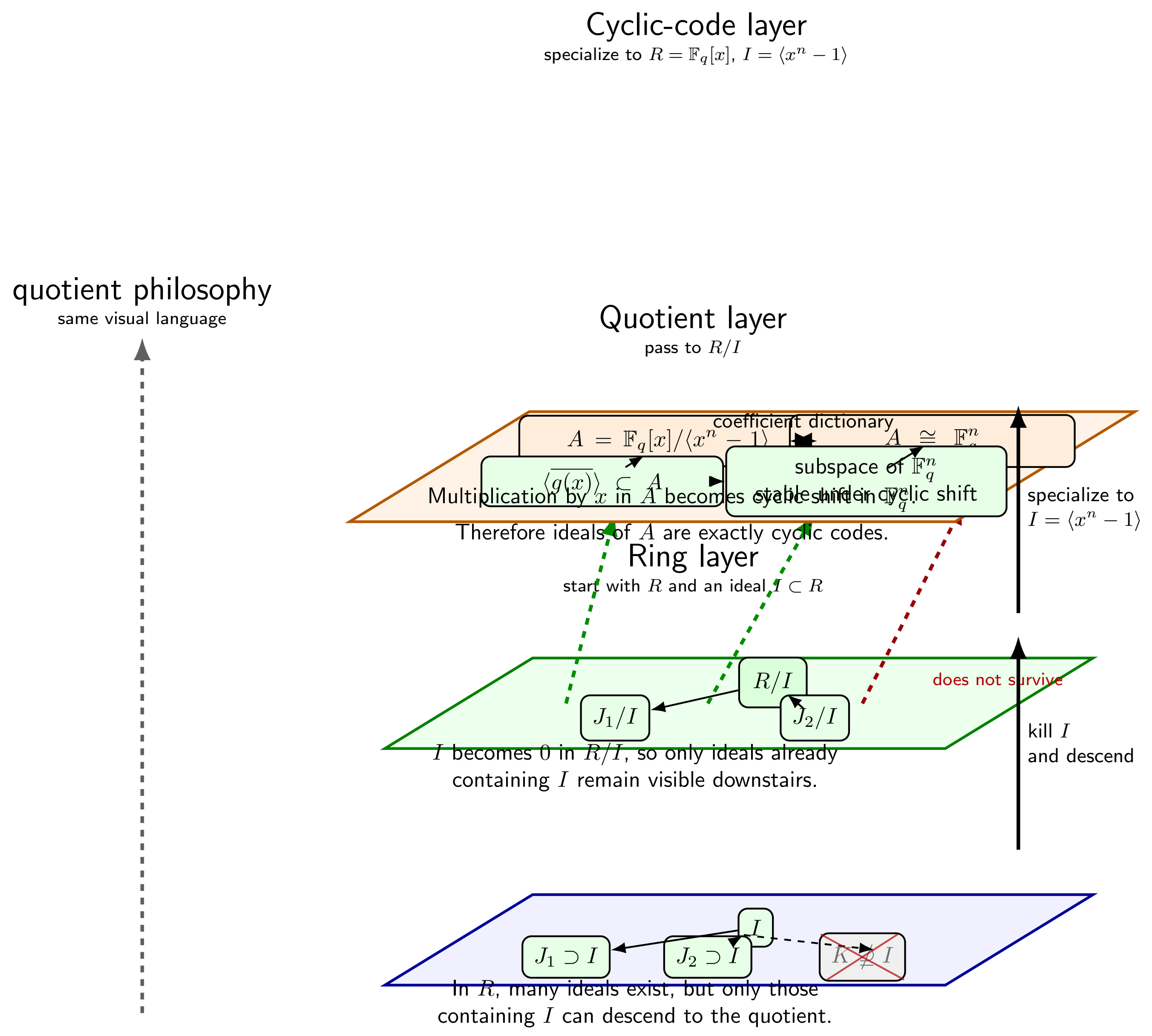
Distributing the message terms:

$$\begin{aligned} c(x) = m_0 &\cdot [g(x)] \\ + m_1 &\cdot [x \cdot g(x)] \\ + m_2 &\cdot [x^2 \cdot g(x)] \\ + m_3 &\cdot [x^3 \cdot g(x)] \end{aligned}$$

Linear Algebra

$$\mathbf{c} = \mathbf{m}G$$

$$\begin{bmatrix} m_0 & m_1 & m_2 & m_3 \end{bmatrix} \begin{array}{r} \phi(g) \\ \phi(xg) \\ \phi(x^2 g) \\ \phi(x^3 g) \end{array} \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} & 0 \\ 0 & 0 & 0 & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$



Passing from R to R/I forces I to become 0, so only ideals containing I survive.

In $A = \mathbb{F}_q[x]/\langle x^n - 1\rangle \cong \mathbb{F}_q^n$, multiplication by x becomes cyclic shift.

Hence ring-theoretic ideals in A = subspaces stable under cyclic shift = cyclic codes.